

Dung Bui

dungbui9736@gmail.com · Department of Computer Science, Northwestern University, Evanston, IL, USA

Education

Université Paris Cité / CNRS Paris, France
PhD in Computer Science, IRIF; Advisor: Geoffroy Couteau Sept. 2021 – Mar. 2025

Université de Limoges Limoges, France
MSc in Mathematics, Cryptology, Coding and Applications (CRYPTIS); ranked 1st Sept. 2019 – July 2021

École Normale Supérieure de Hanoi Hanoi, Vietnam
BSc in Pure Mathematics; highest honours, top 5% Sept. 2015 – July 2019

Thesis: Efficient Secure Computation from Correlated Pseudorandomness; degree awarded 18/03/2025.

Experience

Postdoctoral Researcher Evanston, IL
Northwestern University (host: Xiao Wang) July 2026 – present

Postdoctoral Researcher Paris, France
LIP6, Sorbonne Université / CNRS (hosts: E. Diamanti, A. B. Grilo, L. Perret) June 2025 – June 2026

Visiting Researcher Tokyo, Japan
NTT Social Informatics Laboratories (host: Masayuki Abe) July 2024 – Sept. 2024

Visiting Researcher Leuven, Belgium
COSIC, KU Leuven (host: Nigel Smart) June 2023 – July 2023

Doctoral Researcher Paris, France
IRIF, Université Paris Cité / CNRS (advisor: Geoffroy Couteau) Sept. 2021 – Mar. 2025

Research Intern Paris, France
IRIF, Université Paris Cité (advisor: Geoffroy Couteau) Mar. 2021 – Aug. 2021

Research Intern Limoges, France
XLIM, Université de Limoges (advisor: Duong Hieu Phan) June 2020 – Aug. 2020

Journal articles

1. An Efficient ZK Compiler from SIMD Circuits to General Circuits
Dung Bui, Haotian Chu, Geoffroy Couteau, Xiao Wang, Chenkai Weng, Kang Yang, Yu Yu
Journal of Cryptology (JoC), 2024

Articles in refereed conferences

1. Practical Adaptor Signatures for NP from Online/Offline NIZK
Masayuki Abe, **Dung Bui**, Kelong Cong, Miyako Ohkubo, Zehua Shang, Akira Takahashi
International Conference on the Theory and Application of Cryptology and Information Security (ASIACRYPT), 2026

2. Critical Round in Multi-Round Proofs: Compositions and Transformation to Trapdoor Commitments
Masayuki Abe, David Balbás, **Dung Bui**, Miyako Ohkubo, Zehua Shang, Akira Takahashi, Mehdi Tibouchi
International Conference on the Theory and Applications of Cryptographic Techniques (EUROCRYPT), 2026
3. New Framework for Structure-Aware PSI from Distributed Function Secret Sharing
Dung Bui, Gayathri Garimella, Peihan Miao, Phuoc Van Long Pham
International Conference on the Theory and Application of Cryptology and Information Security (ASIACRYPT), 2025
4. Efficient Fuzzy Labeled PSI from Vector Ring-OLE **(Best Paper Award)**
Dung Bui, Kelong Cong
International Conference on Cryptology and Network Security (CANS), 2025
5. Structured-Seed Local Pseudorandom Generators and Their Applications
Benny Applebaum, **Dung Bui**, Geoffroy Couteau, Nikolas Melissaris
International Conference on Randomization and Computation (RANDOM), 2025
6. Improved All-but-One Vector Commitment with Applications to Post-Quantum Signatures
Dung Bui, Kelong Cong, Cyprien Delpech de Saint Guilhem
Australasian Conference on Information Security and Privacy (ACISP), 2025
7. Efficient Multi-Instance Vector Commitment and Application to Post-Quantum Signatures
Dung Bui
Australasian Conference on Information Security and Privacy (ACISP), 2025
8. Faster Signatures from MPC-in-the-Head
Dung Bui, Eliana Carozza, Geoffroy Couteau, Dahmun Goudarzi, Antoine Joux
International Conference on the Theory and Application of Cryptology and Information Security (ASIACRYPT), 2024
9. FOLEAGE: $\mathbb{F}_4$ OLE-Based Multi-Party Computation for Boolean Circuits
Maxime Bombar, **Dung Bui**, Geoffroy Couteau, Alain Couvreur, Clément Ducros, Sacha Servan-Schreiber
International Conference on the Theory and Application of Cryptology and Information Security (ASIACRYPT), 2024
10. Fast Public-Key Silent OT and More from Constrained Naor-Reingold
Dung Bui, Geoffroy Couteau, Pierre Meyer, Alain Passelègue, Mahshid Riahinia
International Conference on the Theory and Applications of Cryptographic Techniques (EUROCRYPT), 2024
11. Improved Private Set Intersection for Sets with Small Entries
Dung Bui, Geoffroy Couteau
International Conference on Practice and Theory of Public-Key Cryptography (PKC), 2023

Manuscripts

1. Critical-Round Special Soundness for Multi-Round Proofs
Masayuki Abe, David Balbás, **Dung Bui**, Miyako Ohkubo, Zehua Shang, Akira Takahashi, Mehdi Tibouchi
Cryptography ePrint Archive, Report 2026/1679
2. Post-Quantum Anamorphic Signatures: Private, Broadcast, and Threshold Channels from Multi-Round Proofs
Dung Bui, Francesco Migliaro

In submission, 2026

Grants, Awards & Fellowships

1. Honourable Mention, PhD Thesis Prize of GDR Sécurité Informatique (CNRS), 2025 — the French national annual award for a doctoral thesis in information security.
2. Best Paper Award, CANS 2025, for *Efficient Fuzzy Labeled PSI from Vector Ring-OLE*.
3. Fully funded doctoral grant, DIM Math Innov / FSMP, Île-de-France Region, 2021–2025 (42 months).
4. Vingroup Scholarship Programme (30 000 EUR), full scholarship for the CRYPTIS MSc, 2019–2021.
5. VIASM Scholarship (3 000 EUR), National Programme for the Development of Mathematics, Vietnam, 2017–2019.
6. Silver Medal, NSUCRYPTO International Olympiad in Cryptography, 2021.
7. Bronze Medal, Vietnam National Mathematical Olympiad (VMO), 2015.

Teaching & Supervision

Teaching Assistant and Co-instructor

Evanston, IL

Northwestern University

Fall 2026

- COMP_SCI 307: Introduction to Cryptography (undergraduate)

Research Supervision

Evanston, IL

Fengke Li (BSc, Shanghai Jiao Tong University), co-supervised with Xiao Wang July 2026 – Dec. 2026

- Research internship on adaptor signatures from ML-DSA.

Invited Talks

1. May 2026, Apr. 2026, Mar. 2026, Jan. 2026 — *Critical Round in Multi-Round Zero-Knowledge Proofs*, at VIASM (Hanoi, virtual); ALMASTY seminar, Sorbonne Université; Cybersécurité DVRC/ESILV workshop, Paris; INRIA C2 seminar, Paris.
2. Nov. 2025, Apr. 2025, Dec. 2024, Nov. 2024 — *Optimized MPC-in-the-Head Based Signatures from Puncturable PRF*, at NTT Social Informatics Laboratories, Tokyo; VIASM (Hanoi, virtual); Crypto Student Seminar, CWI Amsterdam; Crypto Day, Télécom Paris.
3. Sept. 2025, Mar. 2025 — *FOLEAGE: $\mathbb{F}_4$ OLE-Based MPC for Boolean Circuits*, at Crypto Café, Florida Atlantic University (virtual); ALMASTY seminar, Sorbonne Université.
4. May 2024 ($\times 2$) — *Fast Public-Key Silent OT and More*, at AlgoCRYPT seminar, J.P. Morgan, New York (virtual); Algorithms and Complexity seminar, IRIF, Paris.
5. June 2024 — *Efficient MPC from Correlated Randomness*, at Crypto Seminar, NTT Social Informatics Laboratories, Tokyo.
6. July 2023, May 2023, Apr. 2023, July 2022 — *Efficient PSI from Pseudorandom Correlation Generators*, at COSIC, KU Leuven; VIASM (Hanoi, virtual); Algorithms and Complexity seminar, IRIF, Paris; Crypto Student Seminar, CWI Amsterdam (virtual).

Conference presentations of accepted papers: CANS 2025 (Osaka), ACISP 2025 (Wollongong), ASIACRYPT 2024 (Kolkata), PKC 2023 (Atlanta), Journées C2 2022 (Hendaye).

Service

Programme committee member: CSF 2027, APKC 2026, CCS 2026, ASIACRYPT Artifacts 2026, APKC 2025, LATINCRYPT 2025.

External reviewer: CRYPTO 2025–26; EUROCRYPT 2025–26; ASIACRYPT 2025; PKC 2026; ICALP 2026; CCS 2025; ACNS 2025; PROVSEC 2025; TCC 2022; CSF 2022–23; IEEE Transactions on Information Forensics and Security 2023.

Organiser: IRIF Cryptography Seminar, Université Paris Cité, 2024.

Outreach

- Member, Advanced Cryptography Research Lab, a joint initiative of VIASM and the Vietnam Government Information Security Commission, 2025– (<https://acr-lab.github.io/>).
- Featured interview, *Meet the Women Experts in Quantum Science*, QICS, Sorbonne Université, 2025.
- Co-organizer of QATS, an educational board game on quantum mechanics, and activity leader at *Lumière sur la quantique* (Quantum Day), 2025.
- Featured interview, *Tuoi Tre* (Vietnamese national magazine), 2019.
- Founder and author of the Vcryptis blog for students at the Université de Limoges, 2019–2021 (<http://vcryptis.tech.blog/>).